

Universitat Politècnica de Catalunya

Analysis and Prediction of Apparent cement consumption in Spain

ANÀLISI DE DADES

Data Science and Engineering

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1 Introduction

Cement consumption is an important indicator of economic activity in Spain, as it is directly related to the construction sector and infrastructure development. Monitoring how this variable changes over time can help governments, companies, and researchers understand the current state of the economy and plan for the future. One key event that strongly affected the Spanish economy was the 2008 financial crisis, which caused a sharp drop in construction activity. In this project, we aim to study how this event influenced the time series of cement consumption and whether it is still possible to make accurate forecasts using a statistical model.

The main objective of this project is to estimate a theoretical mathematical model, specifically an ARIMA model that reflects the underlying properties of the data and allows for accurate forecasting of future cement consumption. To build this model, we will follow the Box-Jenkins methodology, which includes a number of structured steps.

Firstly, an initial exploratory analysis will be conducted to describe the series, identifying temporal patterns, cycles, trends, outliers and abrupt changes. Secondly, the original series will be transformed, if necessary, to meet the assumptions required for ARIMA modeling, that is, stationarity. Then, potential ARIMA models will be identified and the parameters of the proposed models will be estimated. Once the models are estimated, we continue with validation, which involves checking whether the residuals (the differences between the predicted and actual values) behave like white noise. This means they should be random, independent, and normally distributed. If the residuals show this behavior, it means the model fits the data well. Finally, we use the best model to forecast cement consumption for the 12 months after the last available observation. These forecasts include confidence intervals to reflect the uncertainty in the predictions.

In addition to the basic ARIMA approach, we will also include extensions such as outlier detection, adjustment for calendar effects (like holidays and working days), and intervention analysis to model the long-term effects of the 2008 crisis. These improvements will help the model better represent real-world events and make the forecasts more reliable.

Note that most of the Figures that are commented during the report appear in the Appendix, which corresponds to the last pages of this file.

2 Basic description of the dataset

The dataset `cemento.dat` contains the apparent consumption in Spain, measured in thousands of tonnes, recorded monthly from 1988 to 2018. However, the analysis excludes data from the first two years. The dataset contains 30 annual observations, corresponding to each year from 1990 to 2019, and the variable represents the total cement consumed in Spain each year.

It is important to note that the data is heavily influenced by the housing crisis, which had a major impact on the country between 2008 and 2014, and can be analyzed visually by examining the time series plot.

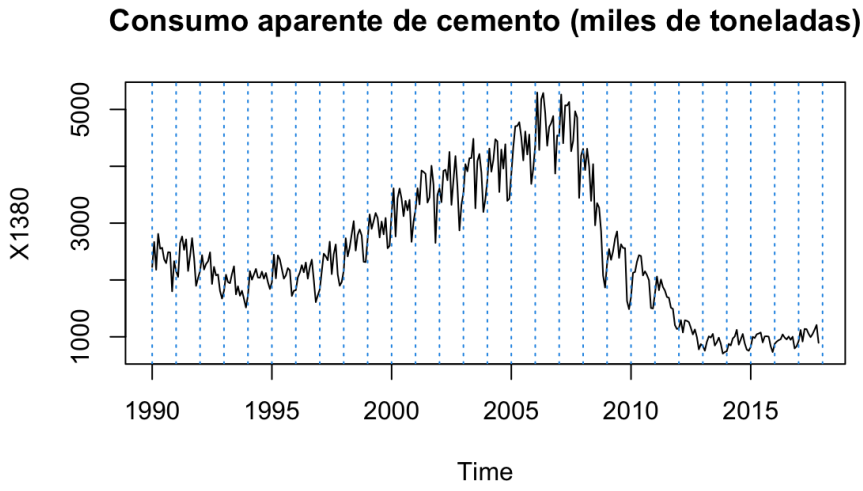


Figura 1: Cement consumption in Spain in thousands of tonnes from 1990 to 2008.

As shown in Figure 1, from the early 1990s until 2007, there is a clear upward trend in cement consumption, reflecting a period of sustained economic growth, construction booms, and rapid urban development in Spain. The increase was especially steep during the early 2000s, aligned with a real estate bubble driven by low interest rates, speculative investment, and high demand for housing and infrastructure.

However, a dramatic decline in cement consumption begins around 2008, coinciding with the global financial crisis. In Spain, the bursting of the housing bubble had a particularly severe impact. The construction sector, which had become a cornerstone of the Spanish economy, collapsed. As a result, cement consumption plummeted sharply from its 2007 peak. This drop continued for several years, indicating a prolonged recession in the construction industry. The decline was not just immediate but also long-lasting. As a result, construction activity decreased drastically, as well as employment and the industry entered a deep decline that lasted until 2014, when we can find the lowest point.

Starting around 2014–2015, the chart starts to increase slowly again, showing a very modest recovery. However, even by 2019, consumption levels remain significantly below pre-crisis highs, suggesting a structural shift in the economy and a more cautious, less construction-dependent growth model.

3 Results and interpretation

3.1 Description and Transformation

To build predictive models, we must transform the series into a stationary form. A time series is considered stationary if it satisfies three key conditions: constant mean, constant variance and autocorrelation matrix independent of the starting point.

The first constrain we need to analyze and transform if necessary is the **constant variance**. By plotting the boxplot (Figure A.1) it was observed that the variance of the series was not constant over time, indicating heteroscedasticity. In particular, during the years of greatest expansion, the variability between the peaks is higher than in 2015 and the following years, where the difference between fluctuations is smaller. Therefore it becomes evident that the variance is not constant, and to stabilize the variance, a logarithmic transformation was applied to the original series. The log-transformed model becomes as it follows in the Figure:

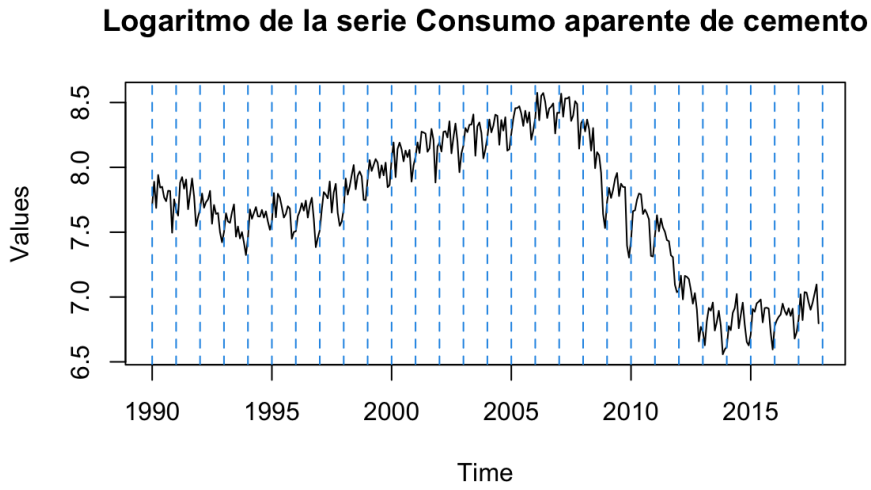


Figura 2: Cemento series log-transformed

Secondly, construction activity throughout the year is not constant. Some months show higher peaks than others. Specifically, November and December are the months with the lowest average of cement consumption while June has the highest average. We can relate this behavior to the weather and the daylight hours. Therefore, it is evident that seasonal pattern exists, and not only can be detected through visual inspection but also confirmed using a `monthplot` (Figure A.2), revealing the presence of periodic fluctuations. To address this, a seasonal differencing was performed.

The final check involves examining the mean of the series, and ensuring that is constant throughout the whole time series. In order to do so we will apply one and two regular differences (Figure A.5 and A.6). With each transformation, the variance decreases significantly up until the last one (as it can be seen in the appendix). This second first-order differencing increases the variance, so we consider that the most appropriate choice is to keep the model with one seasonal difference and one regular difference. If we choosed the model with two regular differences we would be overdifferenciating.

Finally, the resultant stationary series would be:

$$\boxed{W_t = (1-B)(1-B)^{12} \cdot \log(X_t)}$$

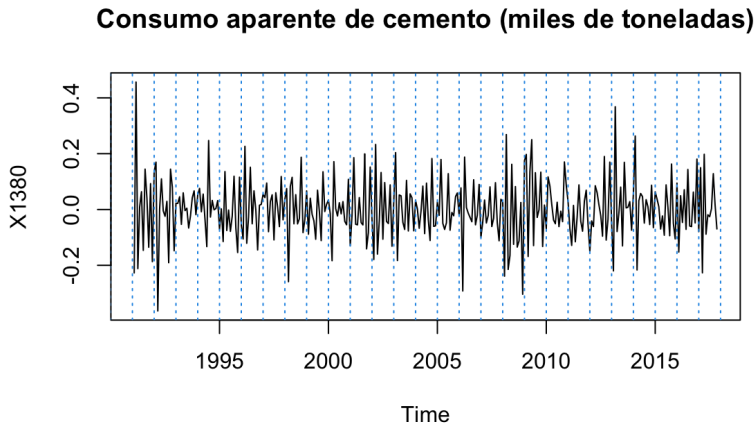


Figura 3: Stationary Transformed Series W_t

3.2 Model Identification

Once the series has been successfully transformed into a stationary one, the next step in the Box-Jenkins methodology is model identification. This involves examining the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots in Figure A.8 to determine the appropriate structure for both the regular and seasonal components of the ARIMA model.

Firstly, we analyze the **regular part**. Since the data follows a seasonal pattern, it is normal to observe **satellite lags**. Therefore, to find the correct parameters, we should focus only on the samples closest to the origin. From the ACF, we identify an **MA(1)** model because there are 1 sample that stand out, and we must consider that significant values start at lag 1. Regarding the PACF, we observe an **AR(2)** model since there are 2 samples that stand out, and in this case, the graph starts at lag 0.

Therefore, we have two models: MA(1) on one side and AR(2) on the other.

Secondly, we analyze the **seasonal part**. To identify the parameters of this part, we focus only on the **samples highlighted in red**, which indicate the seasonality of the series. If we look at the ACF, we see that the first 2 samples stand out from the confidence interval, meaning we have an $MA(2)$ model. Regarding the PACF, we observe an **AR(0)** model.

Thus, we have two simple seasonal models: $MA(2)$ and $AR(0)$.
We also propose the $ARMA(1,1)$ model since there are values of $p = 2$ and $q = 2$ in the regular and seasonal components.
Therefore, we propose the following models:

- Model 1: $ARIMA(2,1,0)(0,1,2)_{12}$
- Model 2: $ARIMA(0,1,1)(0,1,2)_{12}$
- Model 3: $ARIMA(2,1,1)(0,1,2)_{12}$
- Model 4: $ARIMA(2,1,1)(1,1,1)_{12}$

These candidate models are selected on the basis of the behavior observed for both the regular and seasonal components. All models include a regular differencing of order 1 and a seasonal differencing of order 1 to ensure stationarity, as established in the previous step. The seasonal period is 12, which corresponds to the monthly frequency of the time series.

3.3 Estimation

The next step involves estimating the parameters of each model and evaluating their performance. This will be done using standard model selection criteria such as the AIC and BIC and the residual diagnostics. By comparing these metrics across the proposed models, we can determine which specification best captures the underlying dynamics of the time series and ensures white-noise residuals.
After analyzing the parameters of the three models proposed we can see that all of them are significative according to the t-ratio, that is, all of them have:

$$|\hat{t}| = |\frac{coeficient}{\text{s.e coeficient}}| > 2$$

Since all models have significative parameters the next step is to decide which of them is the best one based on adequacy metrics, such as AIC and BIC.

Model	AIC	BIC
m1	-755.7045	-736.8318
m2	-733.2165	-718.1183
m3	-763.1102	-740.4629
m4	-755.6282	-732.9809

Taula 1: Values of AIC and BIC for the models m1, m2 and m3

Table 1 presents the AIC and BIC values for the three models analyzed. We will keep the two ones that present the best properties. In this case:

- **Model 3:** $ARIMA(2,1,1)(0,1,2)_{12} =$

$$(1-B)(1-B^{12})(1-\phi_1 B-\phi_2 B^2)X_t = (1+\theta_1 B)(1+\Theta_1 B^{12}+\Theta_2 B^{2^{12}})Z_t$$

- **Model 4:** $ARIMA(2,1,1)(1,1,1)_{12} =$

$$(1-B)(1-B^{12})(1-\phi_1 B-\phi_2 B^2)(1-\Phi B^{12})X_t = (1-\theta_1 B)(1-\Theta_1 B^{12})Z_t$$

The first candidate to analyze is Model 3, as it presents the lowest AIC and BIC values among all models considered. The second candidate is Model 4, which, although it exhibits slightly higher AIC and BIC values, remains very close in performance to the best model.

3.4 Validation

Furthermore, in order to choose between the two best models, a residual analysis is conducted as part of the time series validation process. This involved examining the residuals of both Model 3 and Model 4 to assess their adequacy in capturing the underlying structure of the data. Specifically, there are three constrains analyzed: **homocedasticity, normality and independence of residuals.**

Validation of Model 3 can be visually analyzed in the Appendix.

Examining the residual plot of Model 3, we notice that variability is quite constant thorughout the whole series. This suggests that the model presents homocedasticity, which is the presence of constant variance. Once we plot the scatterplot (Figure A.10) the red smoothing line shows a slight upward trend, suggesting that residual variability increases over time. Although this line is not completely horizontal, it is almost straight indicating potential homocedasticity meaning that the model's error variance is stable throughout the time series. To confirm our supposition, we do the studentized Breusch-Pagan test and we get the `p-value = 0.1782`. Therefore we cannot reject the null hypothesis that considers constant variance.

As regards to normality, once we plot the Normal Q-Q Plot (Figure A.11) we can observe that both at the lower and the upper extremes the probability of finding values that deviate from the reference line is higher than usual when the residuals are normal. In addition to the outliers, there is a notable density of values that deviate from the line, and the probability of extreme values is higher than expected for a distribution $N(0,1)$. This suggests volatility and heavy tails. Shapiro test confirms our suppositions, since `p-value = 2.628e-05`, and therefore we can reject the null hypothesis that supposes normality.

Assessing independence of residuals, both ACF and PACF (Figure A.13) denote that a few lags slightly exceed the confidence bands. For this reason, we will analyze the validation of the model's independence through the Ljung-Box p-values (Figure A.14). Not all test results remain below the blue line (which indicates value 0.05), indicating that the residuals are not independent. Therefore, we cannot consider them to follow a white noise pattern.

The model is causal, but not invertible since many roots of the MA polynomial are unitary. The results can be seen in the Appendix.

Validation of Model 4 can be visually analyzed in the Appendix.

In the validation of Model 4, similar results are obtained: Residuals have constant variance, but do not follow a normal distribution. In the ACF and PACF of the residuals there is some lag outstanding the confidence bands and therefore the independence of residuals cannot be confirmed. Ljung-Box test rejects the null hypothesis, and therefore the residuals are not independent and not compatible with white noise. The model is again causal but not invertible due to the presence of unitary roots in the MA part.
In terms of validation, any of the two models selected can be properly validated as white noise following the three constrains analyzed. Only constant variance is assessed. However, in terms of AIC the best model is Model 3, so it will be the one selected to generate long-term forecasts.

Therefore, we select Model 3 $ARIMA(2,1,1)(0,1,2)_{12}$ to continue our analysis of the time series.

3.5 Forecasting

For Model 3, we generated predictions for the next 12 observations of year 2018, as well as the 95% confidence intervals. We can see them in Figures 21 and 22. The resulting prediction has an average confidence interval width of 510.245 .

In addition, we trained another model on a windowed version of the series excluding the last 12 observations in order to compare its predictions with the real values of the series. We also computed the confidence intervals and the performance metrics (RMSE, MAE, RMSPE, MAPE and mean IC).

Model	RMSE	MAE	RMSPE	MAPE	Mean IC
Model A	131.478	111.719	0.11979	0.10458	408.090

Taula 2: Predictive Capability of Model 3

The two models, the one with the full time series and the one with the windowed version, have very similar coefficients. Therefore, the model is consistent.

As the study goes on and get new models, we will perform more forecasts with the regular and windowed series and evaluate their metrics in order to compare them.

3.6 Calendar Effects

In order to achieve more accurate long-term forecasts and enhance the assumptions used in model validation, adjustments have been made to account for the impact of the 2008 financial crisis and calendar-related

effects. These two elements are of particular relevance to the time series we are studying.

It is important to take into account that construction activity tends to be higher on working days, which leads to an increase in cement consumption. Similarly, the presence of Easter Holidays may alter the series during May and April months. This is the reason why we will compare the $ARIMA(2,1,1)(0,1,2)_{12}$ model fitted to the series of logarithmic transformed cement production, incorporating different external regressors: no regressor, Easter effect (vEa), Trading days effect (vTD), and both $(vEa + vTD)$. The goal is to assess the impact of these calendar effects on model fit.

The following table shows the results of this experiments:

Taula 3: Comparison of ARIMA(2,1,1)(0,1,2)[12] Models with External Regressors

Parameter	No Regressors	vEa	vTD	vEa + vTD
ar1	-0.9434	-0.9092	-0.8536	-0.8168
ar2	-0.5404	-0.5418	-0.4792	-0.4971
ma1	0.4062	0.4610	0.4103	0.4839
sma1	-0.6382	-0.6258	-0.6632	-0.6468
sma2	-0.3618	-0.3741	-0.3368	-0.3532
vEa	—	0.0586	—	0.0612
vTD	—	—	-0.0068	-0.0071
Log Likelihood	387.56	396.81	397.84	409.68
AIC	-763.11	-779.62	-781.67	**-803.37**

Introducing the Easter effect significantly improves the model, lowering the AIC and increasing the log-likelihood. This suggests that Easter-related effects do contribute on the model.

When we include only the trading days, the model performs slightly better than the Easter-only model. The coefficient for vTD is small but statistically significant, indicating that trading day effects have a measurable impact.

However, the best model fit in terms of lowest AIC and highest log-likelihood is clearly the model that takes into account both calendar effects. Both regressors retain statistical significance in this model. The improvement in fit across the models confirms that including calendar effects (both holiday and trading day adjustments) enhances the explanatory capacity of the model and its forecasting accuracy.
Therefore, the model that we select to continue with the study is:

$$\boxed{\mathrm{W}_{lin} = W_t - \omega_{TD} D_t - \omega_{EA} EA_t}$$

3.7 Effect of the 2008 Crisis

To capture the impact of the 2008 crisis, an intervention analysis was modeled. This analysis was divided into different phases according to three sections of the big downward trend of the crisis seen in the original time series. During the first two years, 2008 and 2009, a considerable impact was identified, as cement consumption dropped drastically. In the following two years, the intensity of the impact decreased, estimated to be roughly half of that experienced during the initial phase of the crisis. From 2011 to 2017, the impact was considered even lower, due to the slow stabilization of the decline.
After fitting the model taking into account this impact of the 2008 crisis the AIC achieves its lowest value with -803.96.
Once we corrected all this effects, the final linearized time series is:

$$\boxed{\mathrm{W}_{lin} = W_t - \omega_{TD} D_t - \omega_{EA} EA_t - \omega Crisi Crisi_{2008}}$$

Having obtained the linearized time series W_{lin}, we can estimate an ARIMA model using the transformed series. Again, we will apply both one regular difference and one seasonal difference to make the series stationary, and in order to select the best model, we will plot and examine the ACF and the PACF of the

differenced series. We can see the ACF and PACF in Figure 23.

The most adequate model seems to be an $ARIMA(4,1,0)(0,1,1)_{12}$. Once the model is trained, we will also subject this model to validation, ensuring that it satisfies the three constrains.

3.7.1 Validation of model ARIMA (4, 1, 0) (0, 1, 1) [12]

The validation of the model can be seen in Figures 24 to 29. Regarding homocedasticity, the residuals plot and the scatter-plot of absolute residuals give a very similar result to the previous model. The studentized Breusch-Pagan test returns a p-value of 0.3768, therefore we can assume that it has constant variance.
The normal Q-Q plot, as in the previous case, has slight deviations on the tails, suggesting that the distribution has heavy tails and volatility. The normal distribution doesn't completely fit with the residuals histogram, but it seems better than the previous case. The Shapiro results in a p-value of 2.609e-6, which confirms that the residuals might not follow a normal distribution.
Finally, both the ACF and the PACF plots of the residual indicate that there are several lags with values significantly different than zero. On the Ljung-Box plot, we see that the first 8 lags have a significant value. Therefore, we can't assume that the residuals are completely independent.
Since the roots of both AR and MA polynomials have a modulus greater than 1, the model is causal and invertible.

3.7.2 Forecasting with ARIMA (4, 1, 0) (0, 1, 1) [12]

Furthermore, we will also generate the two forecasts. The first, with predictions for the following unknown values, has an average confidence interval width of 498.238, less than the previous one, and it can be seen in Figure 30.

The second forecast, with the same model trained with the windowed series, is shown in Figure 31 and its performance metrics are the following:

Model	RMSE	MAE	RMSPE	MAPE	Mean IC
Model B	70.049	56.012	0.06580	0.05374	459.035

Taula 4: Predictive Capability of Model B, which considers calendar effects and crisis intervention

Unexpectedly, this model has much better RMSE, MAE, RMSPE and MAPE metrics than the previous one, but has a much worse mean IC. We should remember that the first four metrics depend on the point predictions, while the last one depends on the confidence interval. Therefore, our model has better predictions than the last one, but the confidence intervals are worse. When we compare the plot of the predictions of this model and the previous one, we clearly notice that the predictions are nearer the observed values but the interval is wider. In practical terms, the model has accurate predictions but it is less certain about them.

Similar to the previous case, this ARIMA is also consistent because the two models have very similar coefficients.

4 Outlier Treatment

With the previous model adjusted for the calendar and crisis effects, we can finally apply automatic outlier detection. The algorithm finds a total of 9 outliers.

Taula 5: Outliers

Index	Observation	Type detected	W Coef	Abs LRatio	Date	PercVal
4	14	AO	-0.1748515	3.917244	Feb 1991	83.95817
3	221	LS	-0.2047877	3.868675	May 2008	81.48203
5	227	LS	-0.1931048	3.809788	Nov 2008	82.43956
7	228	AO	-0.1379300	3.238314	Dic 2008	87.11597
1	239	TC	-0.2444220	4.343008	Nov 2009	78.31571
6	240	TC	-0.1546758	3.038454	Dic 2009	85.66928
9	272	LS	-0.1463339	3.036968	Ago 2012	86.38692
2	278	AO	-0.1837199	3.895875	Feb 2013	83.21688
8	312	TC	-0.1522941	3.016166	Dic 2015	85.87357

It's important to note that three of the outliers are downwards level shifts on the crisis period, between 2008 and 2014. This suggests that the intervention variables of the crisis may not have fully captured all the anomalous behaviour of the series.

We can proceed subtracting the outliers effect from the previous series, resulting in a last linearized series, free from any exogenous anomalies. Calendar, crisis and outliers effects have been removed.

We can see this linearized series compared with the original one in Figure 32. We can also plot the difference between the original series and the one adjusted from outliers, which allows us to completely isolate the effect from the outliers. Finally, we can also plot the difference between the original series and the completely linearized one, adjusted for calendar effects, intervention variables and outliers. They are shown in Figures 33 and 34.

We will train our last model based on this final linearized series. In order to select the best model, we will analyse the ACF and PACF, shown in Figure 35. Although the ACF and PACF are very similar to the previous ones, we tested and compared some other competent models, and we determined that the one with the best AIC is an ARIMA (2, 1, 1)(2, 1, 0)[12]. Again, we will validate it.

4.0.1 Validation of model ARIMA (2, 1, 1) (2, 1, 0) [12]

This model presents much better validation results, which can be seen in figures 36 to 41.

Regarding homocedasticity, both residuals plots show that the variance is constant. The red line in the scatter-plot has a slight deviation at the beginning, but the overall trend seems practically flat. In addition, the studentized Breusch-Pagan test gives a p-value of 0.683, therefore we can assume constant variance.

Assessing normality, the Normal Q-Q plot residuals are almost perfectly aligned along the straight line, despite a little deviation at the lower tail. The normal distribution fits very well the residuals histogram. The Shapiro test p-value is 0.0498, which is very close to the 0.05 threshold; therefore, we can practically assume that the residuals follow a normal distribution.

While some lags in the ACF and PACF are significantly different from zero, the Ljung-Box test shows that only the first two lags exceed the confidence band. Thus, we cannot completely assume independence of the residuals, although the overall behaviour is reasonably good.

Finally, the model is both causal and invertible, since the modulus of all the roots of all characteristic polynomials are greater than one.

4.0.2 Forecasting with ARIMA (4, 1, 0) (0, 1, 1) [12]

We will generate more forecasts with this model. The one for the future 12 observations results in an average confidence interval width of 452.318, much better than the previous, and the one with the windowed series has the following metrics:

Model	RMSE	MAE	RMSPE	MAPE	Mean IC
Model C	75.824	64.119	0.07436	0.06296	426.602

Taula 6: Predictive Capability of Model C, which considers calendar, crisis and outliers effects

Compared to the previous model, this one has worse RMSE, MAE, RMSPE and MAPE , but a much better mean IC. This means that the point predictions have worsen but the confidence intervals are narrower. Both forecast can be seen in figures 42 and 43.

5 Results and Interpretation

In this section we will compare the evolution of our initial model with the several modifications done during the study of the series in order to compare and evaluate the results. We will distinguish three versions of the model:

- Model A: fitted without any additional treatment. It's an ARIMA(2, 1, 1)(0, 1, 2)[12].
- Model B: with calendar effects and intervention analysis. It's an ARIMA(4, 1, 0)(0, 1, 1)[12].
- Model C: with calendar effects, intervention analysis and outlier detection. It's an ARIMA(2,1,1)(2,1,0)[12].

For each model, we compile all its performance variables in the following table. MeanCI_1 relates to the future forecasts, while RMSE, MAE, RMSPE, MAPE and MeanCI_2 are the metrics from the forecasts of the windowed models, obtained from predicting the last 12 values and comparing them to the real observations. In this sense, we could say that each model is, in fact, two, one trained with the full time series and the other trained with the windowed version excluding the last 12 observations.

Taula 7: Model Performance Comparison

Model	σ^2	AIC	BIC	MeanCI 1	RMSE	MAE	RMSPE	MAPE	MeanCI 2	#params
A	0.00473	-763.110	-740.463	510.245	131.478	111.719	0.11979	0.10458	408.090	5
B	0.00404	-803.956	-762.436	498.238	70.049	56.012	0.06580	0.05374	459.035	10
C	0.00373	-842.436	-766.588	452.318	75.824	64.119	0.07436	0.06296	426.602	10

Now that we have gathered all the information needed, we can begin to draw conclusions.

To begin with, as we move from the first to the third model, the estimated variance σ^2, the AIC and the BIC decrease significantly, indicating that the models are progressively improving. In addition, each model satisfies the validation constraints better than the previous one. The last model is the only one that nearly achieves full validation on all three constraints. In this sense, the first model is the weakest and the third is the best.

On the other hand, while the first model has only five parameters, the other two have ten, so they are much more complex. However, since they have a much lower AIC and BIC and the validation improves, we can deduce that there's not a significant over-fitting.

Regarding the future forecast, we see a similar pattern, as we go on, the prediction intervals get narrower, which also suggests that the third model is the one giving the most accurate forecasts.

When analysing the point-prediction metrics (RMSE, MAE, RMSPE and MAPE) of the forecast for the windowed time series, the first model has the worst metrics and the other two are much better. When we observe the prediction plots carefully, we see that the first model tends to consistently predict lower values. This happens because it was trained with the original time series, which contained many arbitrary and abrupt downward shifts that should have not been taken into account. The other two models, on the other hand, always predict higher and more accurate predictions, since they were trained on linearized series where those downwards shift where removed. In other words, those anomalies where treated as exceptions and not as regular behaviours to be learned. This explains why the first model performs worse than the other two.

Between the second and third model, the performance is quite similar, although model B has slight better metrics. However, when observing the series during the last few years, it becomes clear that the data is significantly random with no clear pattern. Therefore, we believe that the superior performance of the second model is mainly due to random chance rather than a structural difference between the models.

Lastly, when looking at the mean prediction interval for the windowed case, we find an interesting situation: In this case, the first model has the narrower intervals, the second has the widest, and the third falls in between. This is the opposite of what happened with the point-prediction metrics, where the first model was the worse and the second the best.

However, upon closer inspection, we notice that although the first model's intervals are narrower, they aren't quite good. There's an observation that falls outside the intervals and others are almost on the edge. In general, the real observations are often not centred within the intervals. Therefore, even though the model has narrower confidence intervals, they are misleading and not very reliable. In contrast, the other models produce wider intervals but capture the underlying randomness of the behaviour of the series much more accurately. In simpler terms, the first model is "overconfident", it provides very thigh intervals but often misses the actual observations, while the other two are more "honest" and acknowledge the true underlying uncertainty.

In conclusion, Model C appears to be the best overall, achieving the strongest validation, good predictive performance, and providing realistic and trustworthy prediction intervals.

6 Conclusions and discussion

The analysis confirmed the initial hypotheses: the 2008 financial crisis and calendar effects (such as Easter and working days) significantly influenced consumption patterns. Including these factors improved the model's AIC and predicting capabilities, highlighting the importance in explaining the data. The findings support the idea that more working days correlate with higher construction activity, and therefore, greater cement use. The impact of the crisis was strong initially but decreased over time.

This project allowed us to proof that accounting for outliers and calendar effects is crucial, as they can significantly affect model validation. Moreover, understanding the historical and temporal context is essential for accurate predictions and interpretations.

Given the economic relevance of the series, a further improvement of the project could be to use a more advance model that accounts for series volatility.

In conclusion, the project demonstrates the power of the time series analysis to understand and forecast data trends, in this case, the Spanish cement consumption from 1990 to 2018 with strong prediction accuracy beyond the observed data.

7 References

Boletín estadístico online - Información estadística - Ministerio de Fomento. (s. f.).

Lecture Notes on Forecasting Time Series by Josep Anton Sanchez-Espigares and Lesly María Acosta Argueta

8. Annexe

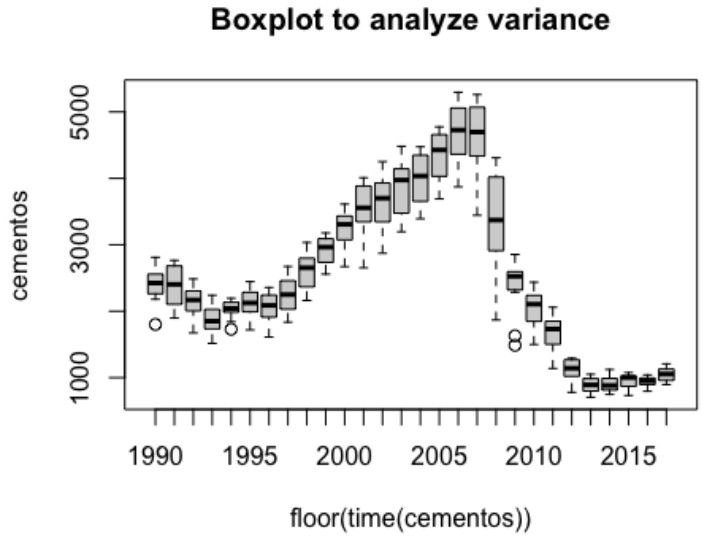


Figure 1: Boxplot of **cementos** to analyze the variance

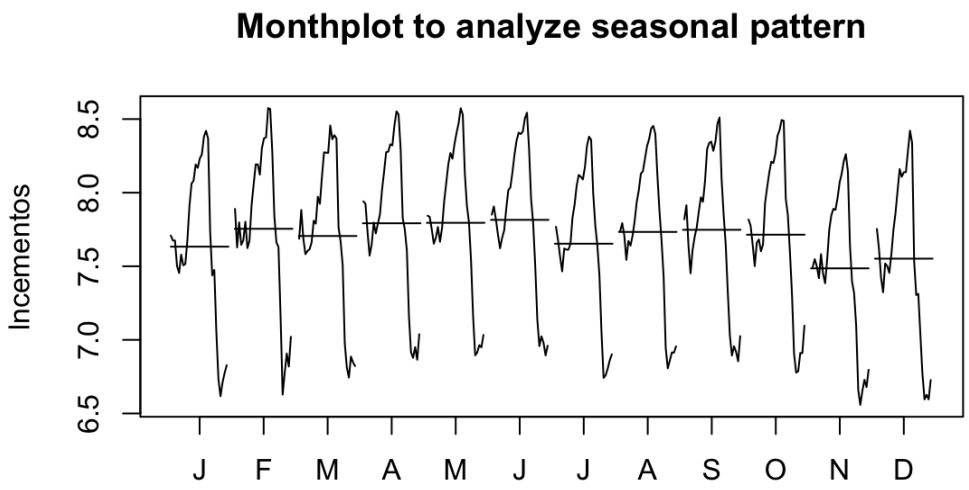


Figure 2: Monthplot of **lncementos** to analyze the seasonal pattern

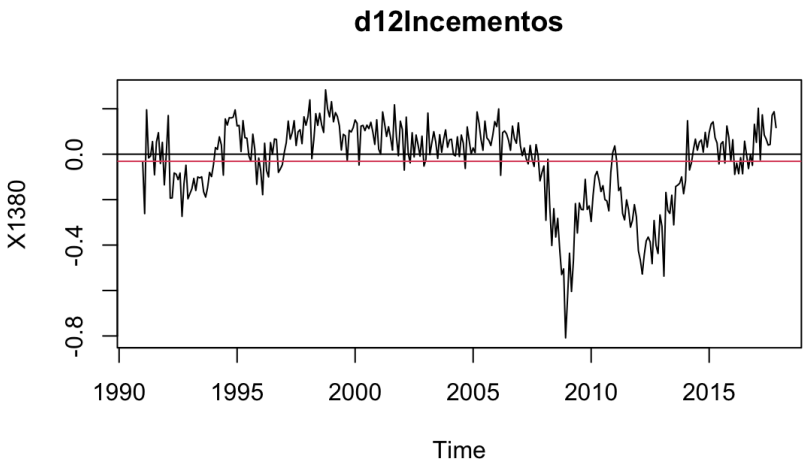


Figure 3: Series **d12lncementos** after seasonal differencing

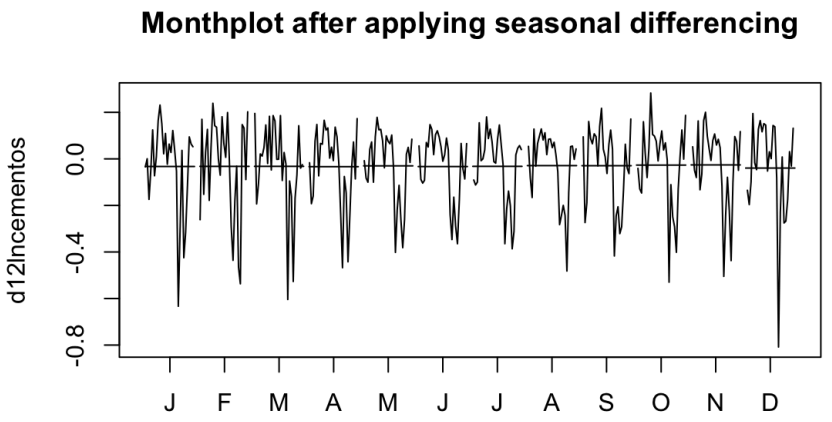


Figure 4: Monthplot of series **d12lncementos** after applying seasonal differencing

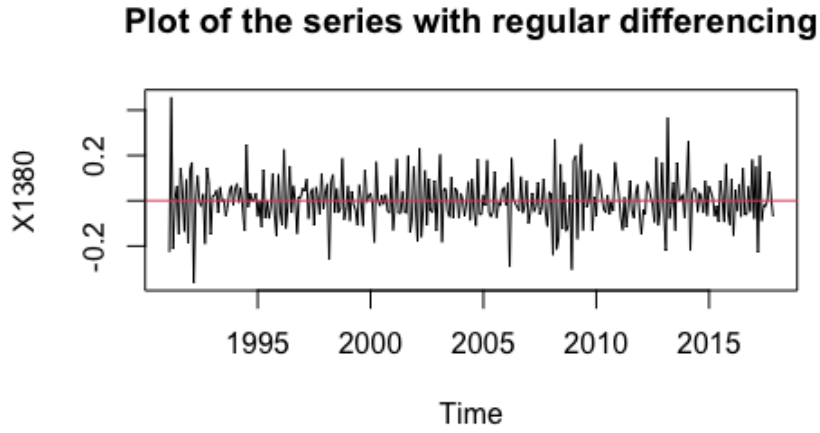


Figure 5: Plot of the series **d1d12lncementos** with one regular differencing

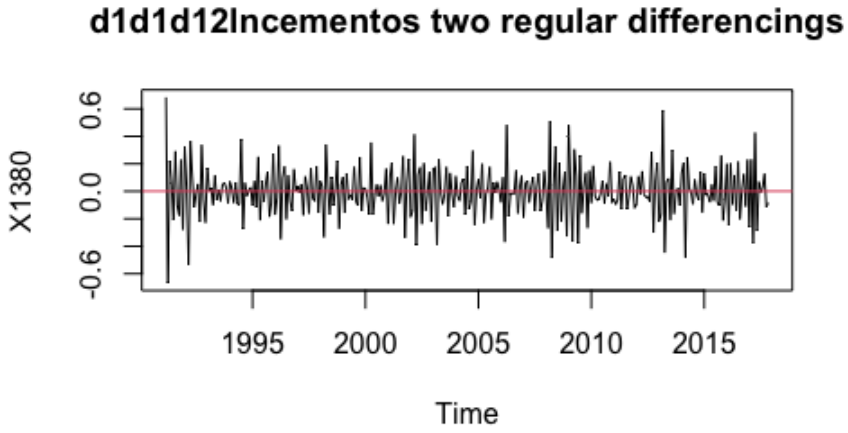


Figure 6: Plot of the series with two regular differencings **d1d1d12lncementos**

```
var lncementos 0.2794664
var d12lncementos 0.03058632
var d1d12lncementos 0.01192622
var d1d1d12lncementos 0.03486676
```

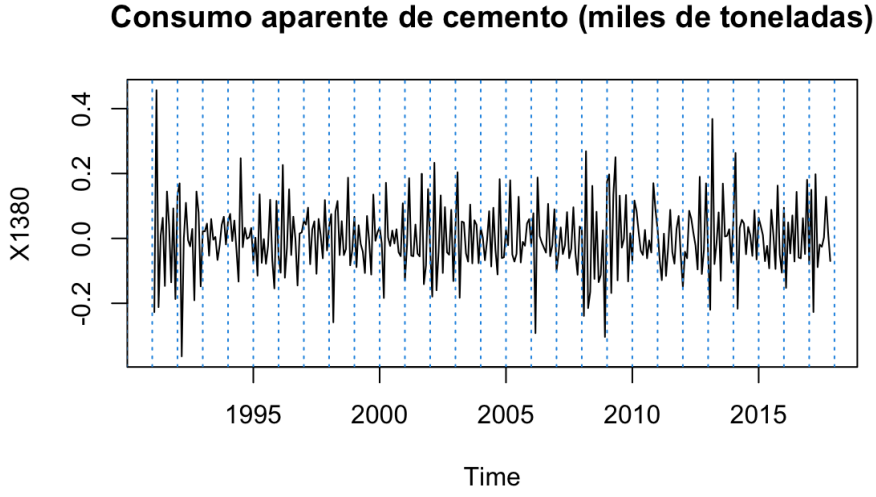


Figure 7: Final transformed selected series: **d1d12lncementos**

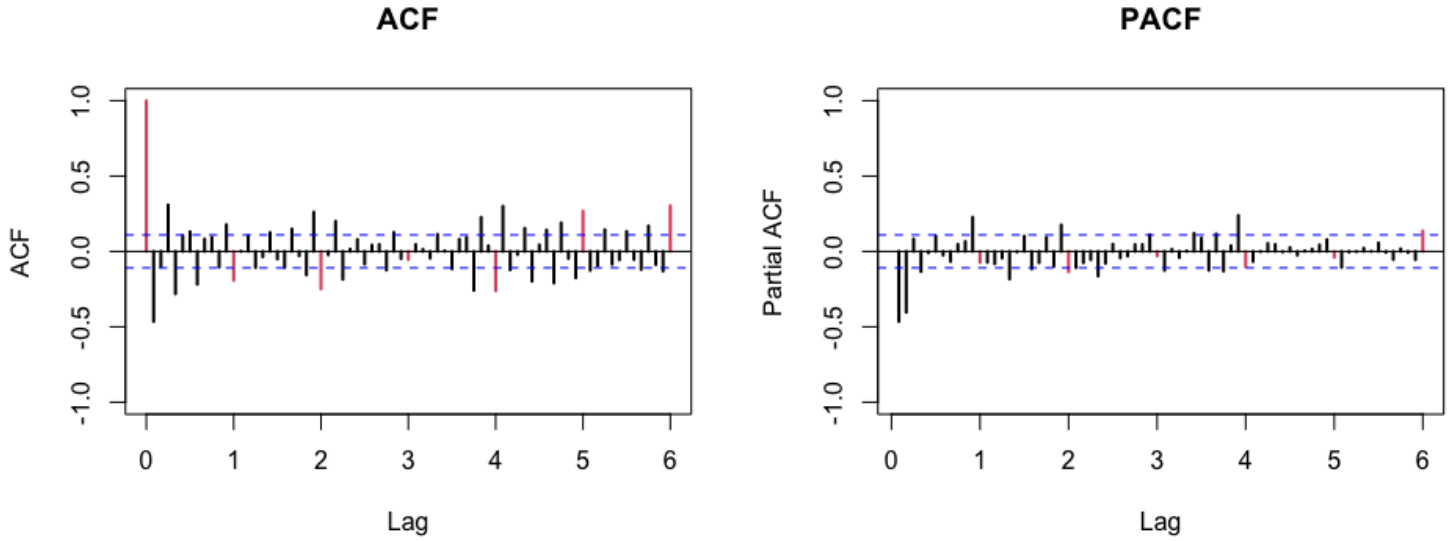


Figure 8: ACF and PACF of **d1d12lncementos**

VALIDATION OF MODEL 3 ARIMA(2,1,1)$(0,1,2)_{12}$

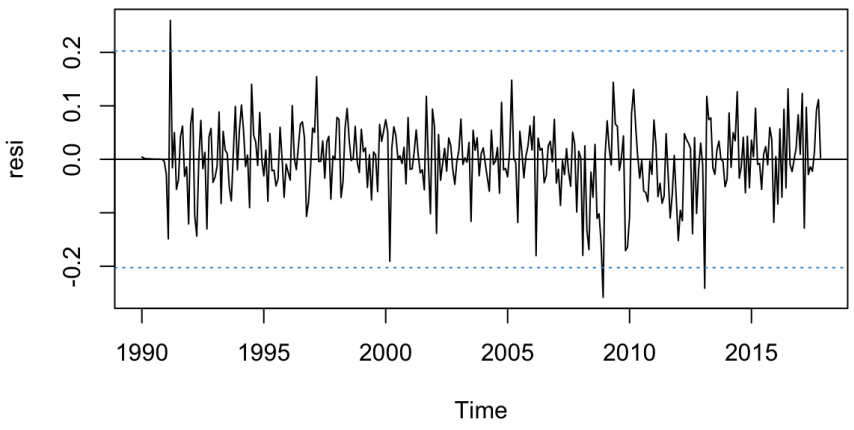


Figure 9: Residual plot of Model 3

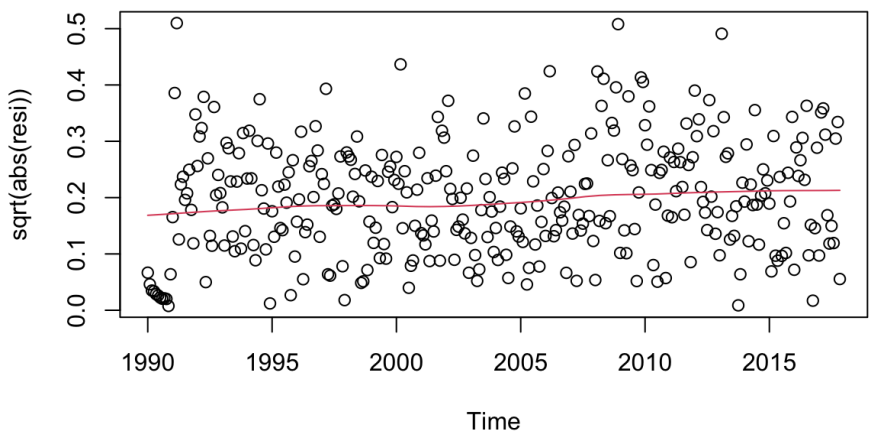


Figure 10: Scatterplot of residuals Model 3

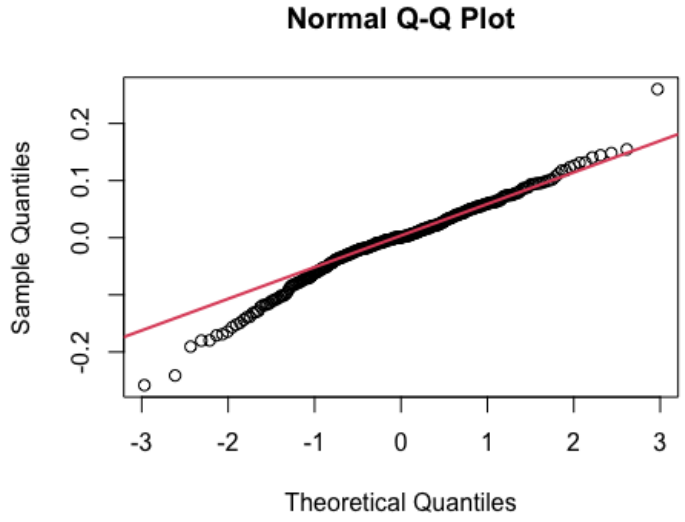


Figure 11: Normal plot residuals of Model 3

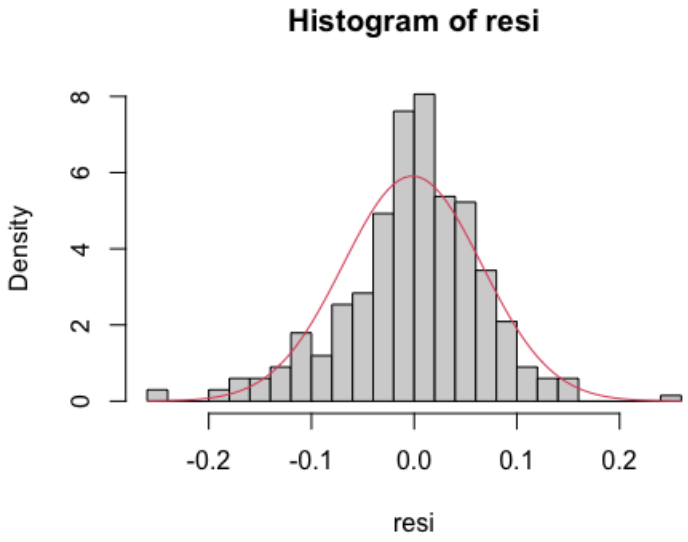


Figure 12: Histogram of residuals of Model 3

```
Shapiro-Wilk normality test

data:  resi
W = 0.97638, p-value = 2.628e-05
```

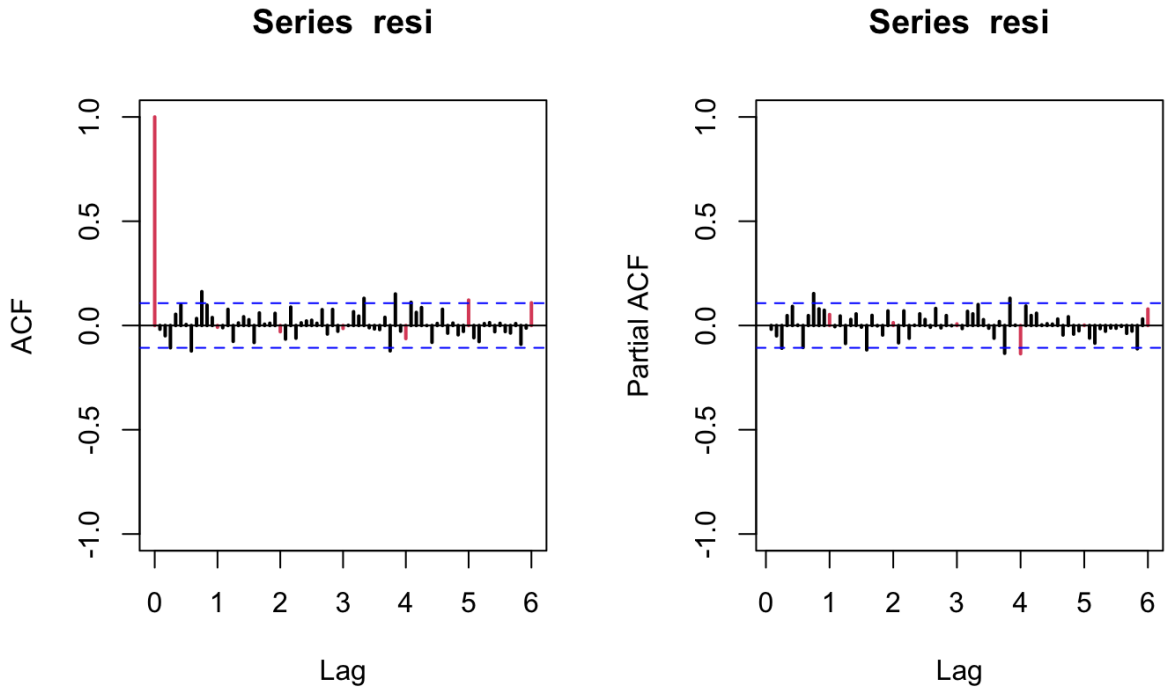


Figure 13: ACF and PACF of the residuals of Model 3

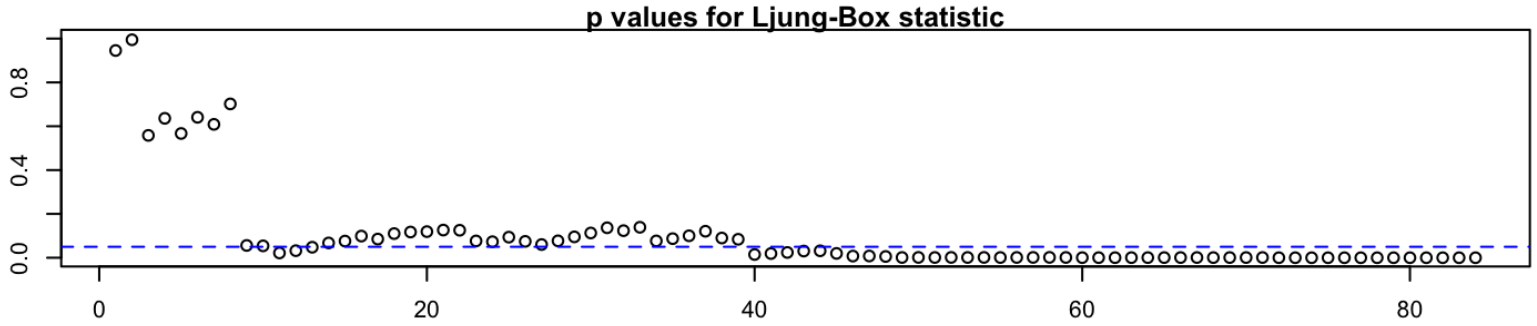


Figure 14: p values for Ljung-Box statistic

```
Mod(polyroot(c(1,m3$model$theta)))

[1] 1.088415 1.000000 1.000000 1.000000 1.000000 1.088415 1.000000
1.000000
 [9] 1.000000 1.000000 1.088415 1.088415 1.000000 1.000000 1.000000
1.000000
[17] 1.088415 1.088415 1.088415 1.088415 1.088415 1.088415 1.088415
1.088415
[25] 2.461886
```

```
Mod(polyroot(c(1,-m3$model$phi)))
[1] 1.360352 1.360352
```

VALIDATION OF MODEL 4 ARIMA(2,1,1)(1,1,1)$_{12}$

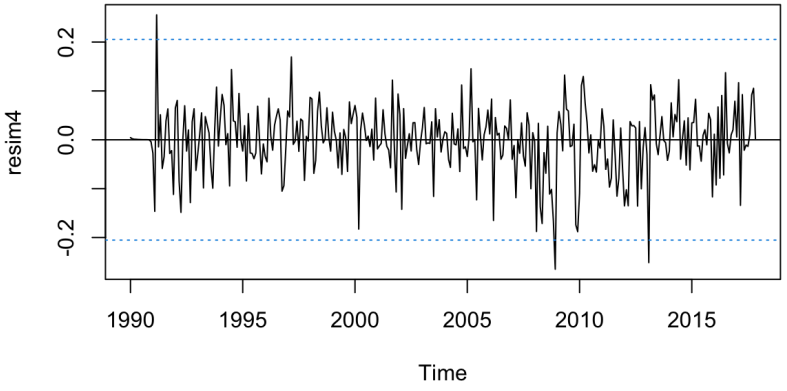


Figure 15: Residuals of Model 4

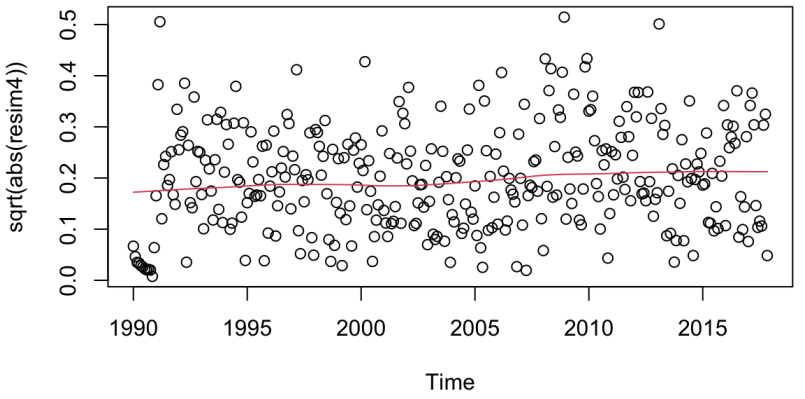


Figure 16: Scatterplot of residuals Model 4

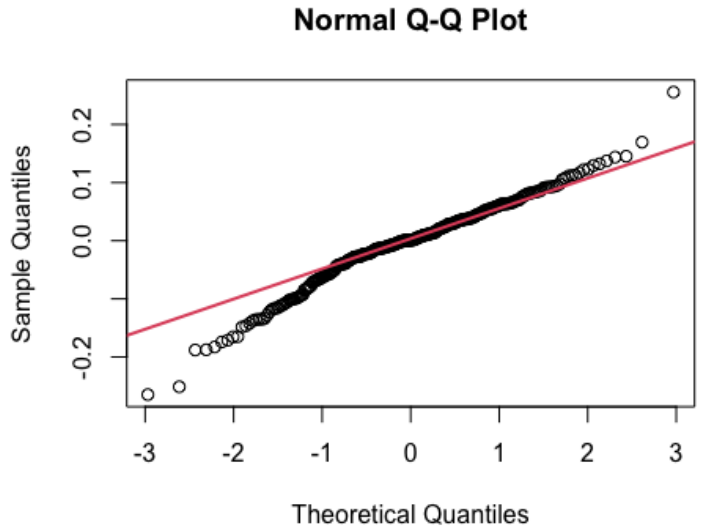


Figure 17: Normal plot residuals of Model 4

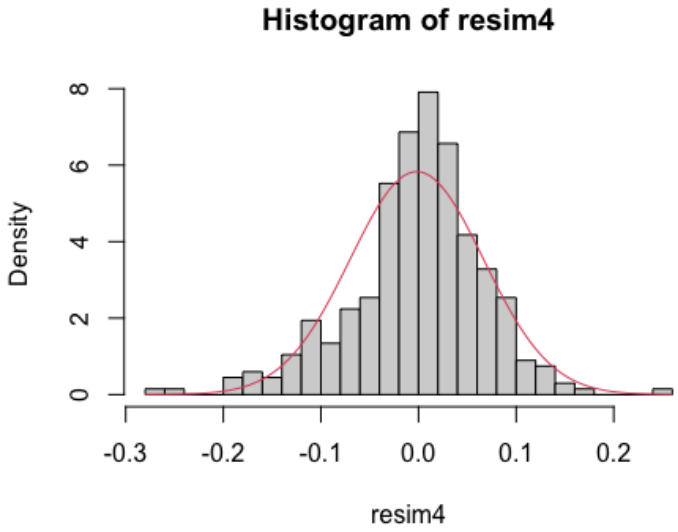


Figure 18: Histogram of residuals of Model 4

```
        Shapiro-Wilk normality test

data:  resim4
W = 0.97329, p-value = 7.302e-06
```

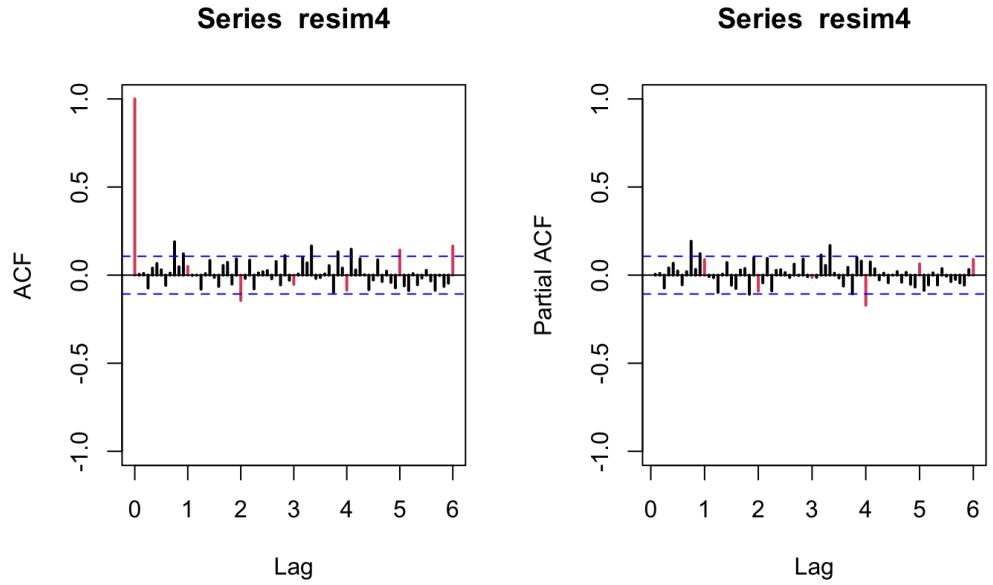


Figure 19: ACF and PACF of the residuals of Model 4

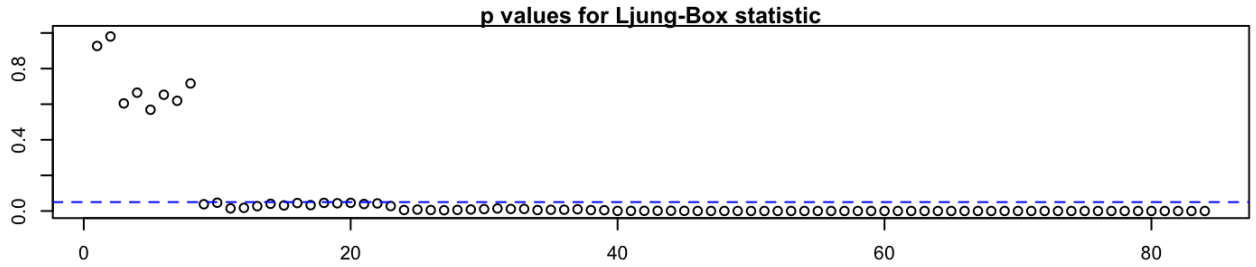


Figure 20: p values for Ljung-Box statistic

```
Mod(polyroot(c(1,m4$model$theta)))
[1] 1.000000 1.000000 1.000000 1.000000 1.000000 1.000000 1.000000
1.000000
 [9] 1.000000 1.000000 1.000000 1.000000 2.443121
```

```
Mod(polyroot(c(1,-m4$model$phi)))
[1] 1.113118 1.113118 1.113118 1.113118 1.113118 1.113118 1.113118
1.113118
 [9] 1.113118 1.113118 1.113118 1.113118 1.348219 1.348219
```

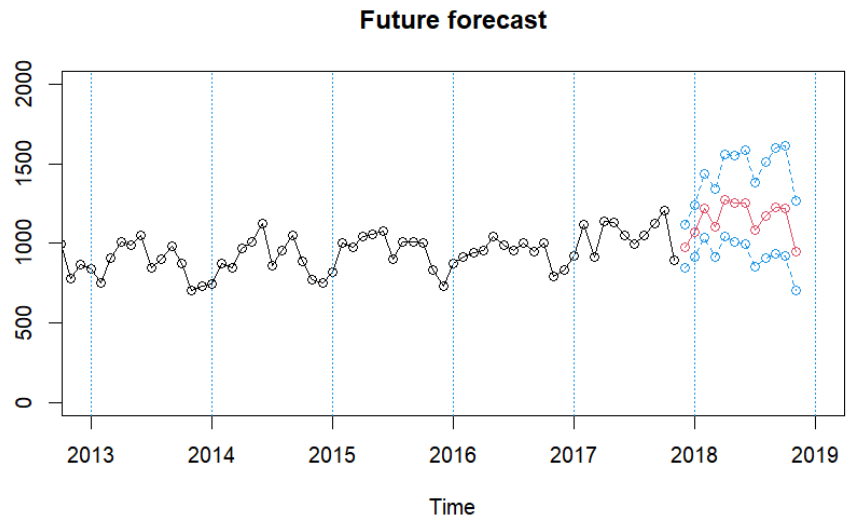


Figure 21, future forecast for Model 3

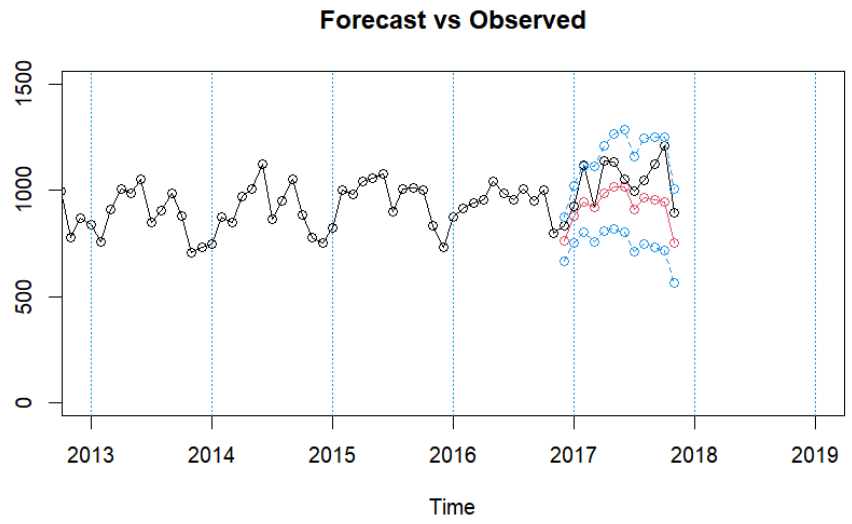


Figure 22, windowed forecast for Model 3

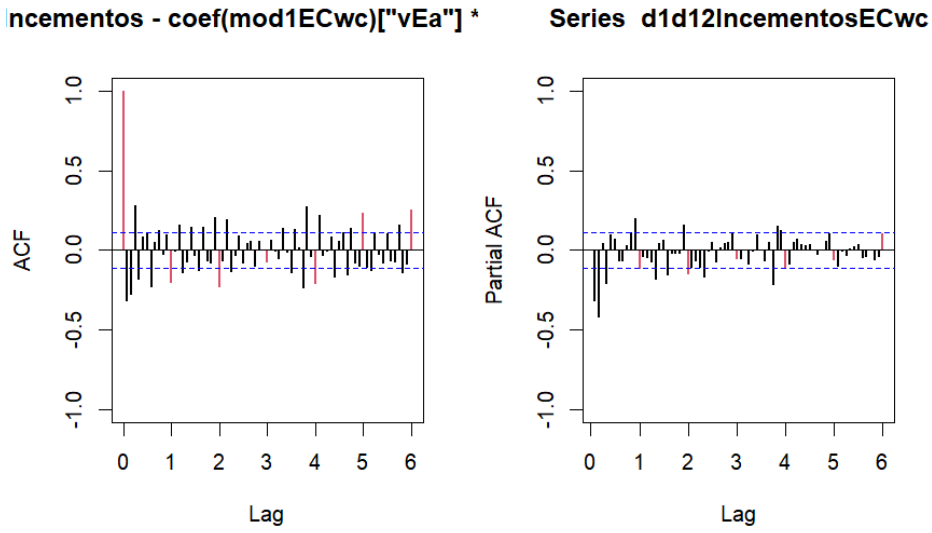


Figure 23: ACF and PACF of the series with calendar and crisis effects removed

VALIDATION OF MODEL B, ARIMA(4, 1, 0)(0, 1, 1)$_{12}$

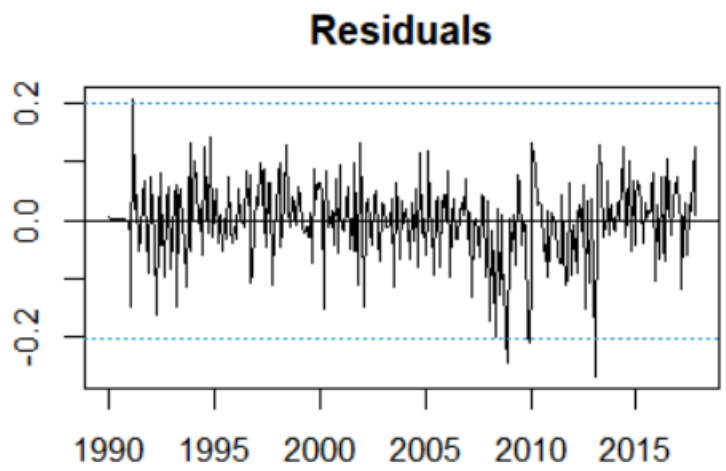


Figure 24: Residuals of Model 4

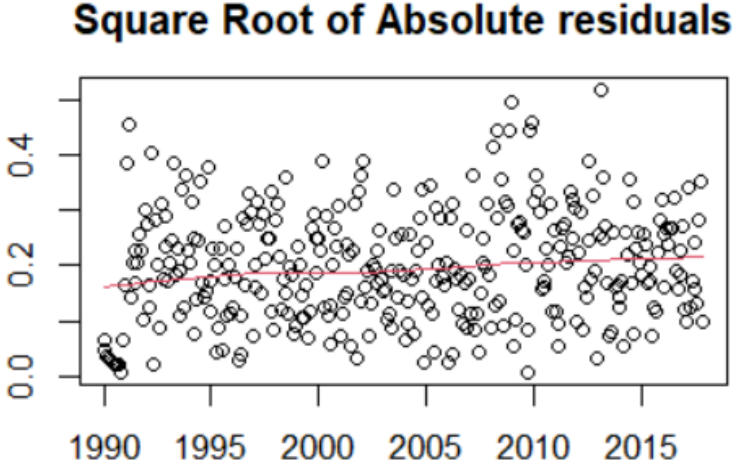


Figure 25: Scatterplot of residuals Model 4

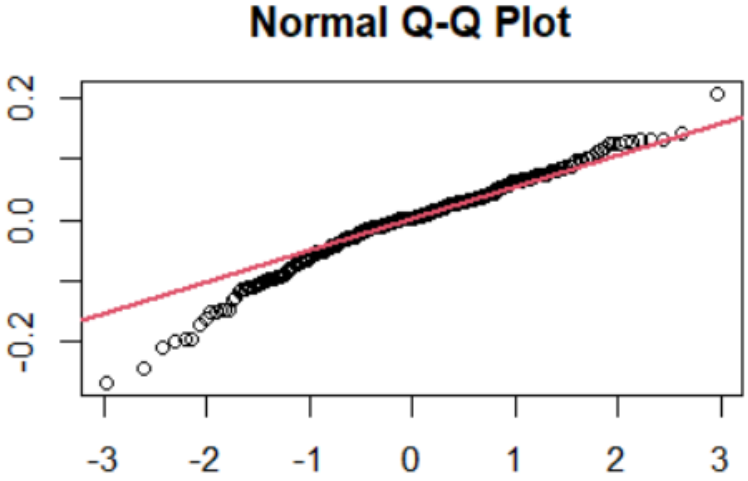


Figure 26: Normal plot residuals of Model 4

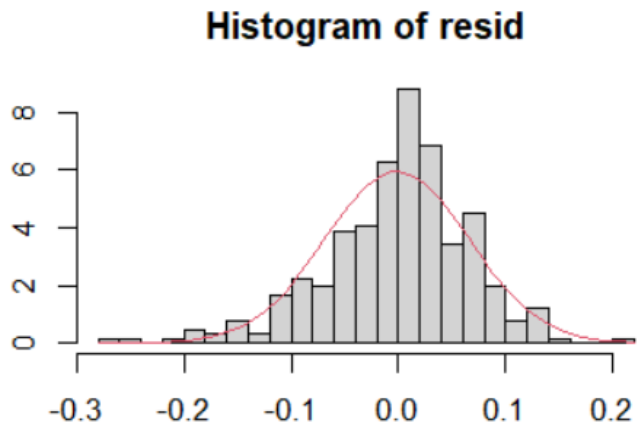


Figure 27: Histogram of residuals of Model 4

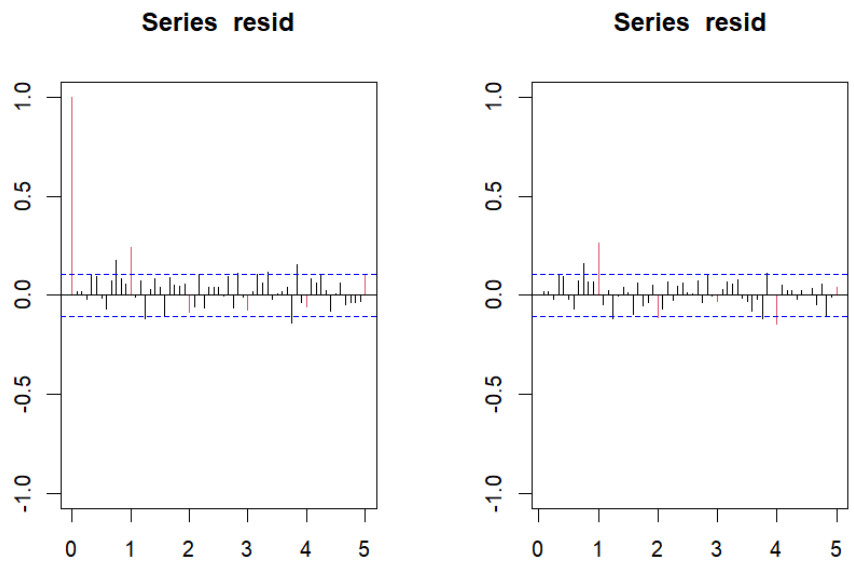


Figure 28: ACF and PACF of the residuals of Model 4

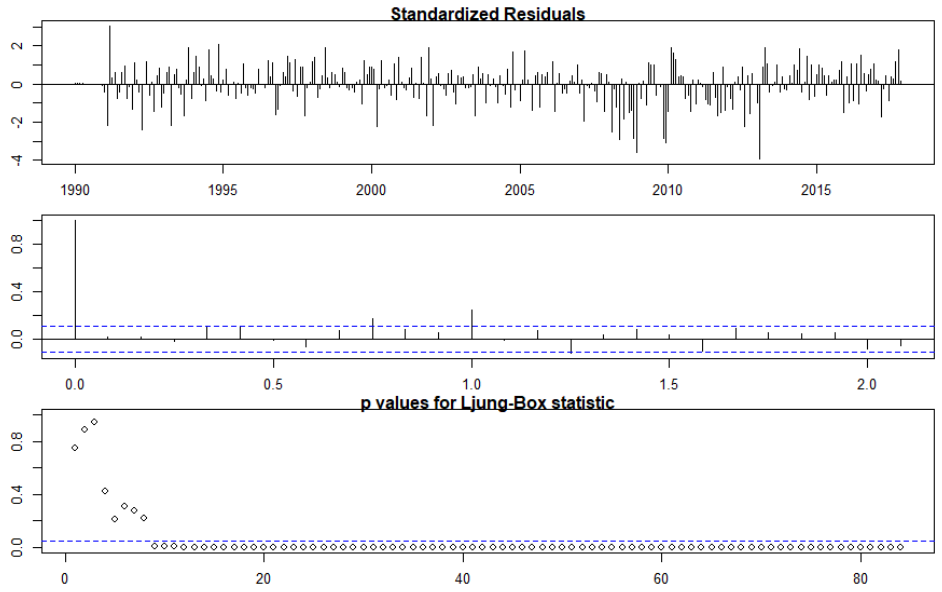


Figure 29: p values for Ljung-Box statistic

```
Modul of AR Characteristic polynomial Roots:  1.881446
1.27988 1.27988 1.881446
```

```
Modul of MA Characteristic polynomial Roots:  1.012028
1.012028 1.012028 1.012028 1.012028 1.012028 1.012028
1.012028 1.012028 1.012028 1.012028 1.012028
```

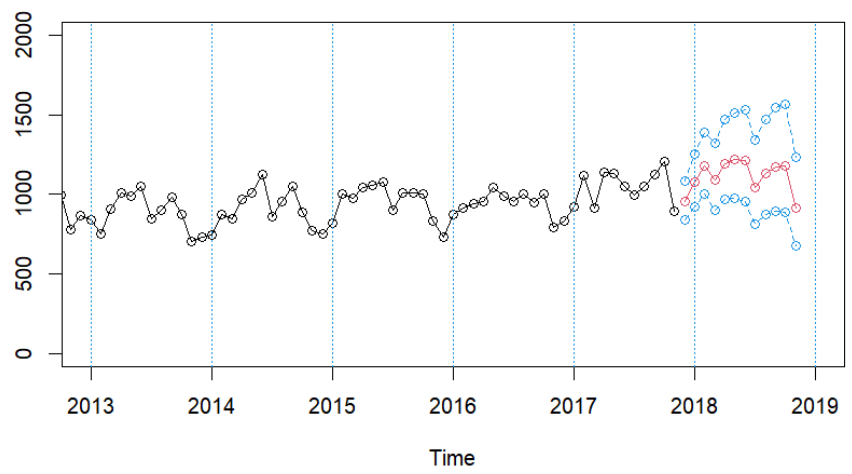


Figure 30: Future forecast for model B

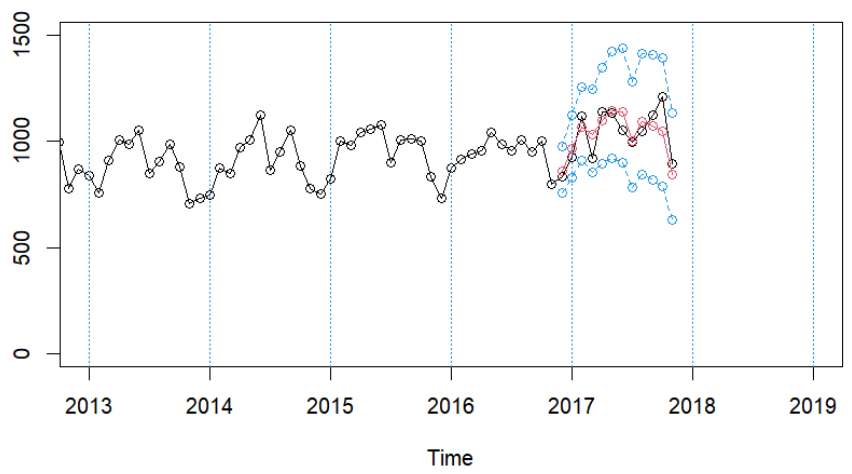


Figure 31: Windowed forecast for model B

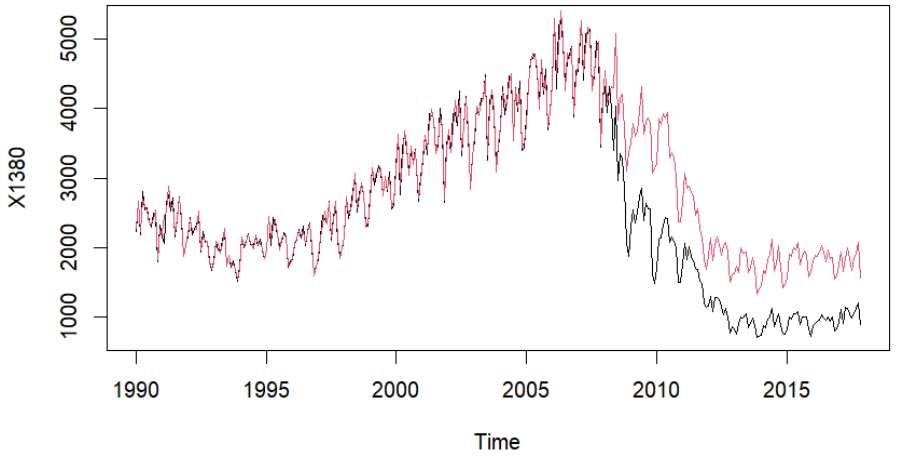


Figure 32: Original series (black) versus fully linearized series (red)

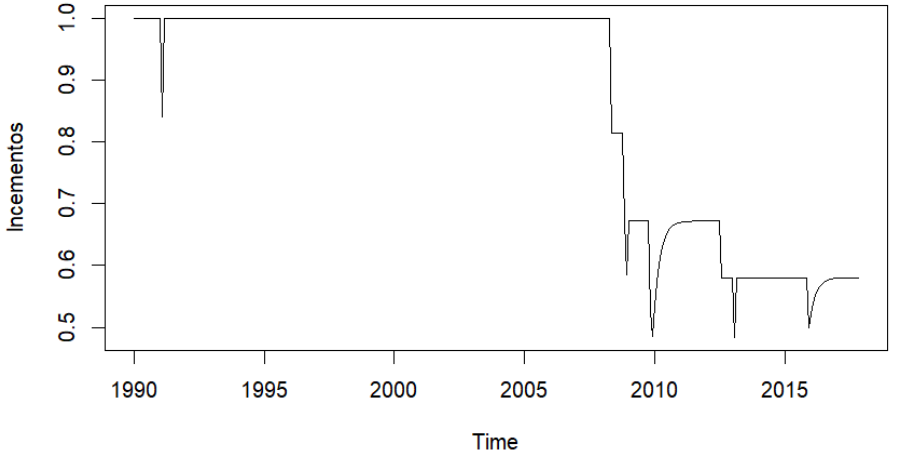


Figure 33: Outlier isolated behavior

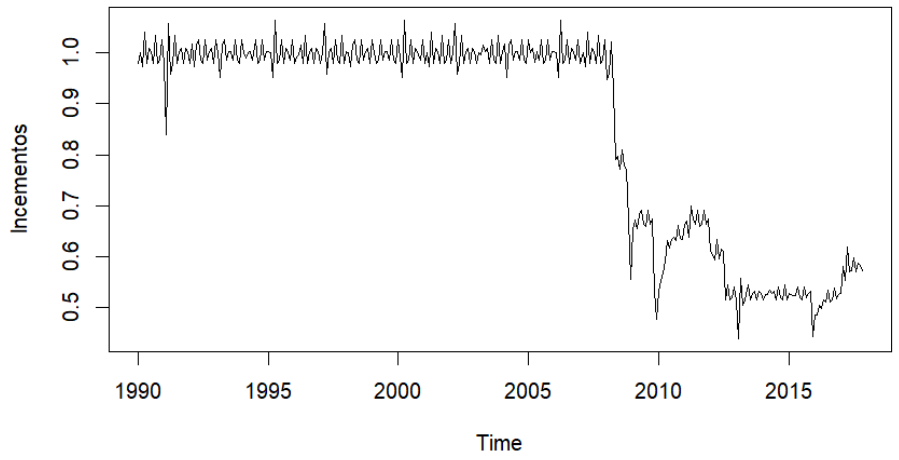


Figure 34: Outlier, calendar and intervention effects isolated

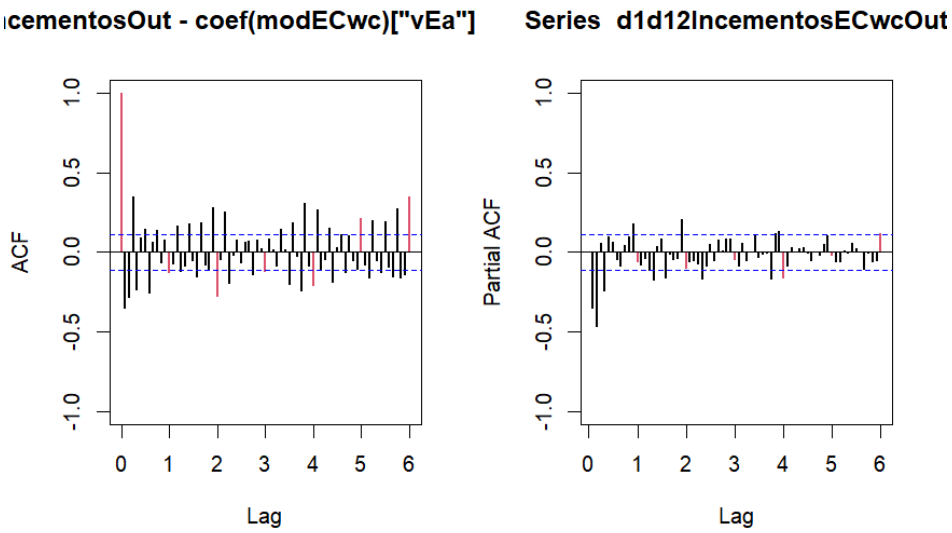
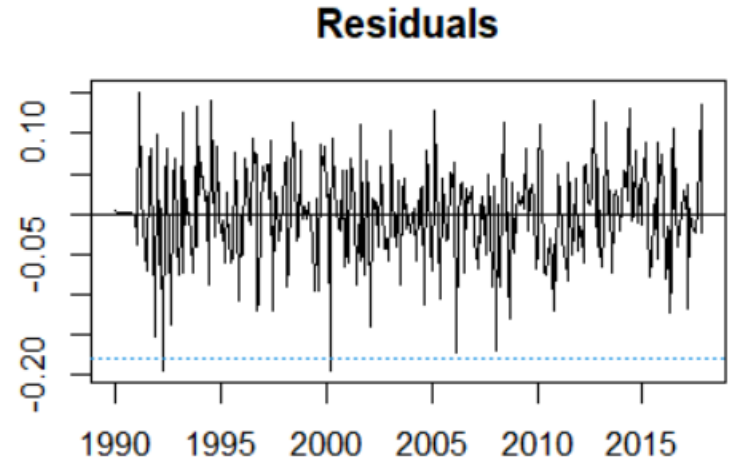


Figure 35: ACF and PACF of the fully linearized series.

VALIDATION OF MODEL C, ARIMA(2, 1, 1)(2, 1, 0)$_{12}$



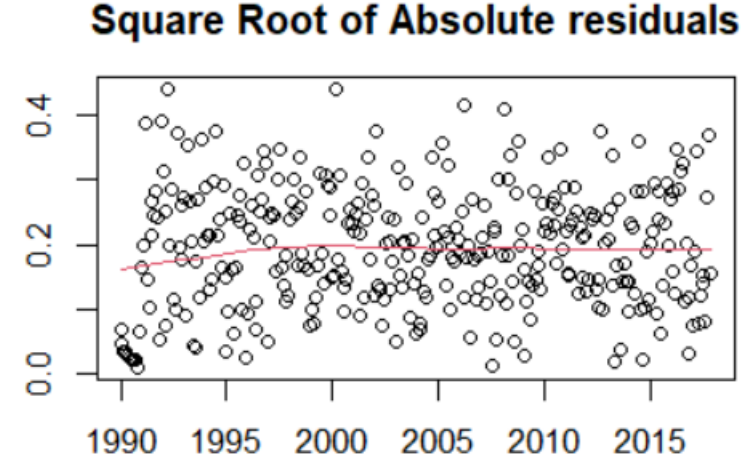


Figure 36: Residuals of Model C

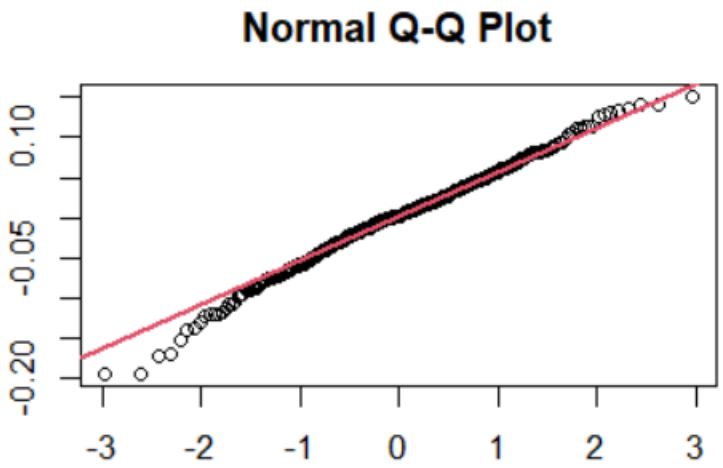


Figure 37: Scatterplot of residuals Model C

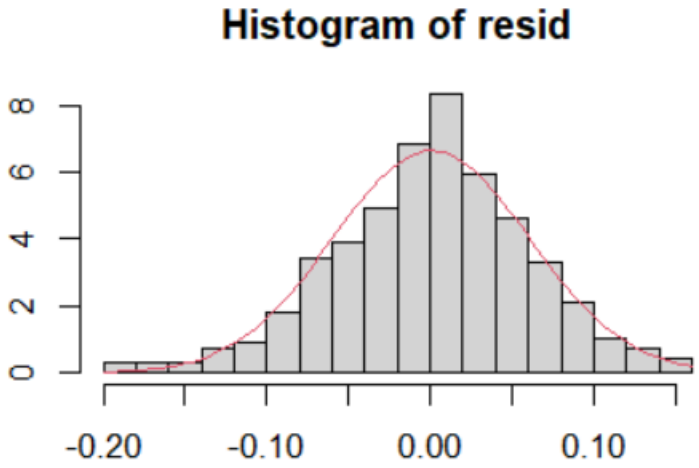
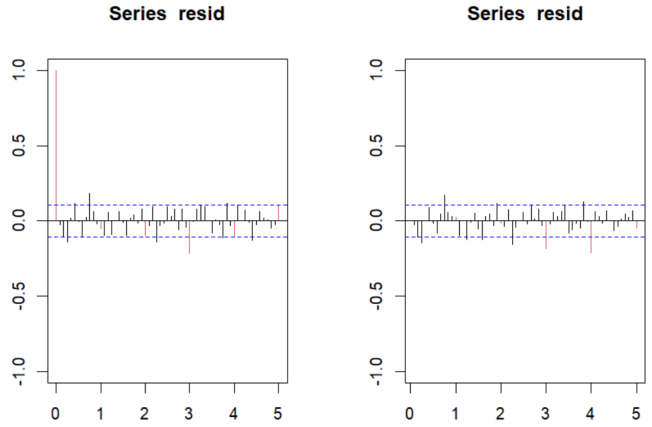


Figure 38: Normal plot residuals of Model C

Figure 39: Histogram of residuals of Model C



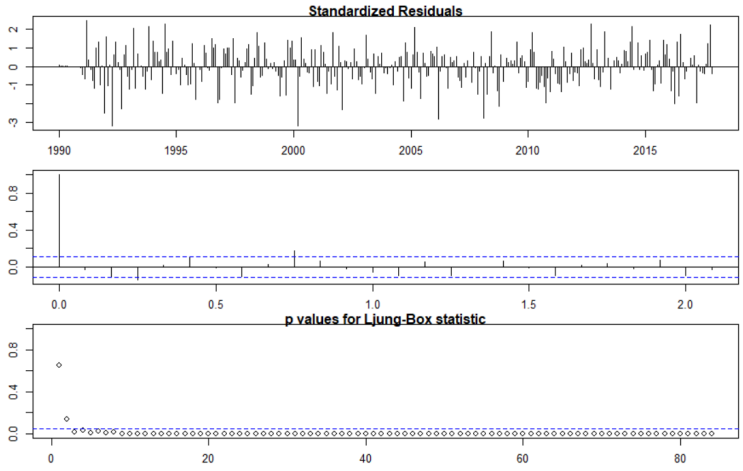


Figure 40: ACF and PACF of the residuals of Model C

Figure 41: p values for Ljung-Box statistic

```
Modul of AR Characteristic polynomial Roots:  1.04769
1.04769 1.04769 1.04769 1.04769 1.04769 1.04769 1.04769
1.04769 1.04769 1.04769 1.04769 1.04769 1.04769 1.04769
1.04769 1.04769 1.04769 1.04769 1.04769 1.04769 1.04769
1.04769 1.296501 1.04769 1.296501
```

```
Modul of MA Characteristic polynomial Roots:  2.076806
```

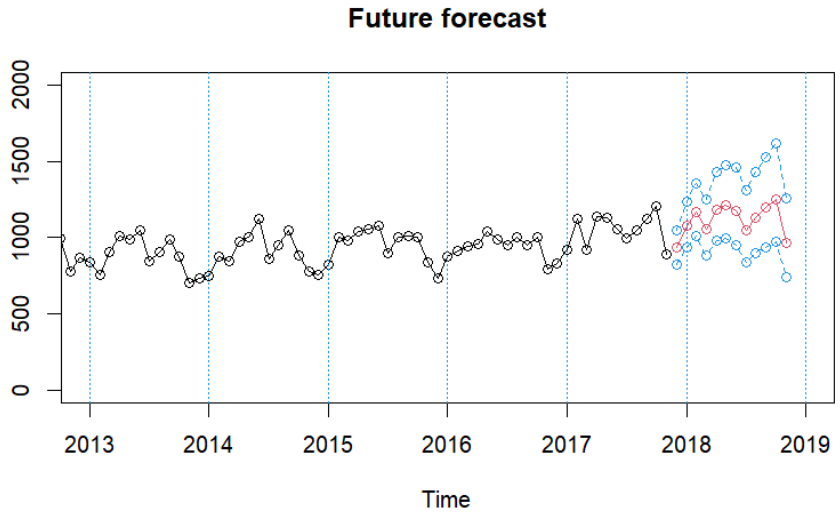


Figure 42: Future forecast for model C

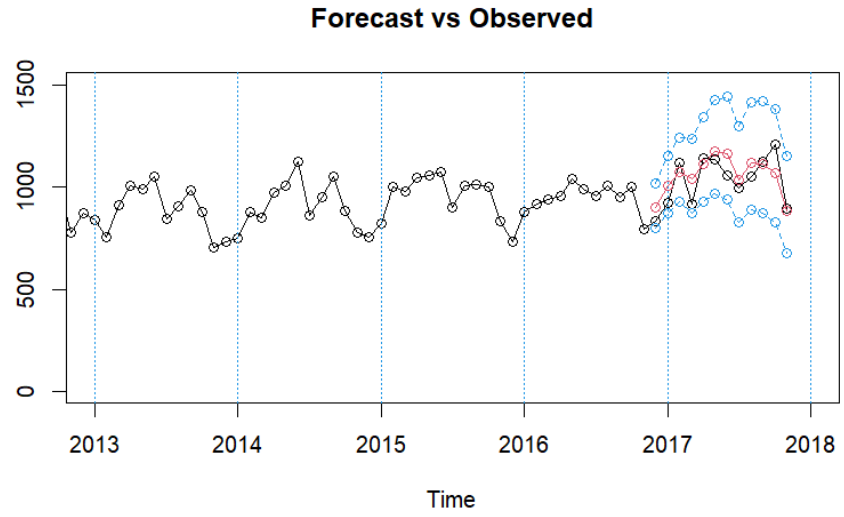


Figure 43: Windowed forecast for model C